

LECTURE 6

EC-241 OOP

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7.7 static Class Members

- **static** class members
 - Shared by all objects of a class
 - Efficient when a single copy of data is enough
 - Only the **static** variable has to be updated
 - May seem like global variables, but have class scope
 - only accessible to objects of same class
 - Exist even if no instances (objects) of the class exist
 - Both variables and functions can be **static**

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7.7 static Class Members

- **static** variables
 - Static variables are accessible through any object of the class
 - **public static** variables
 - Can also be accessed using scope resolution operator (::)

Employee::count
 - **private static** variables
 - When no class member objects exist, can only be accessed via a **public static** member function
 - To call a **public static** member function combine the class name, the :: operator and the function name

Employee::getCount()



7.7 static Class Members

- **Static** functions
 - **static** member functions cannot access non-**static** data or functions
 - There is no **this** pointer for **static** functions, they exist independent of objects



```
#include <iostream>
using namespace std;
class Employee {
private:
    static int count;
    int age;
public:
    static void setCount(int val)
    {
        count=val;
        //age=81;illegal, static functions cannot
        access non-static data
    }
    static int getCount()
    {
        return count;
    }
};
```

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```
int Employee::count=0;//compulsory to initialize
void main()
{
    //cout<<Employee::count; illegal, count is private
    cout<<Employee::getCount();//0
    Employee::setCount(10);
    cout<<Employee::getCount();//10
    Employee e1,e2;
    cout<<e1.getCount();//10
    cout<<e2.getCount();//10
    e1.setCount(20);
    cout<<Employee::getCount();//20
    cout<<e1.getCount();//20
    cout<<e2.getCount();//20
    cout<<endl;
}
```

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